

### **Practice Problems : Chapter 3**

1. A function is given by  $f(x) = x e^{-3x} + x^2$ . Now answer the following up to five significant figures.
  - a. Approximate the derivative of  $f(x)$  at  $x_0 = 2$  with step size  $h = 0.1$  using the central difference method.
  - b. Calculate the upper bound of the truncation error of  $f(x)$  at  $x_0 = 2$  using  $h = 0.1$  using the central difference method.
  - c. Compute  $D_{0.1}^{(1)}$  at  $x_0 = 2$  using Richardson extrapolation method and calculate the relative error.
2. During the derivation process, we explored the first-order Richardson extrapolated derivative using  $(h \rightarrow \frac{h}{2})$ ,
 
$$D_h^{(1)} \equiv f'(x_0) - \frac{h^4}{480} + O(h^6)$$
  - a. Using  $h \rightarrow \frac{h}{2}$ , derive the expression for  $D_{\frac{h}{2}}^{(2)}$ , which represents the second-order Richardson extrapolation.
  - b. Now, starting from the definition of  $D_h^{(1)}$  and applying  $h \rightarrow \frac{h}{3}$ , derive the expression for  $D_h^{(1)}$ .
3. Consider the function  $g(x) = \ln(x)$ ,
  - a. Approximate the derivative of  $g(x)$  at  $x_0 = 1.5$  with step size  $h = 0.2$  using the central difference method up to 6 significant figures.
  - b. Approximate the derivative of  $g(x)$  at  $x_0 = 1.5$  with step size  $h = 0.2$  using the forward difference method up to 6 significant figures.
  - c. Calculate the truncation error of  $g(x)$  at  $x_0 = 1.5$  using  $h = 0.1, 0.01, 0.001$  in both the central difference and forward difference methods.
  - d. Compute  $D_{0.2}^{(1)}$  at  $x_0 = 1.5$  using Richardson extrapolation method up to 6 significant figures and calculate the truncation error.  $D_{0.2}^{(1)}$  at  $x_0 = 1.5$  using Richardson extrapolation method up to 6 significant figures and calculate the truncation error.
4. Given the function  $f(x) = e^{-2x}$ :
  - a. Approximate the derivative of  $f(x)$  at  $x_0 = 0.5$  with step size  $h = 0.2$  using the forward difference method up to 6 significant figures.
  - b. Approximate the derivative of  $f(x)$  at  $x_0 = 0.5$  with step size  $h = 0.2$  using the central difference method up to 6 significant figures.
  - c. Calculate the truncation error of the derivative approximation at  $x_0 = 0.5$  using  $h = 0.1, 0.01, 0.001$  for both the forward and central difference methods.
  - d. Compute  $D_{0.2}^{(1)}$  at  $x_0 = 0.5$  using Richardson extrapolation up to 6 significant figures and calculate the truncation/relative error.

5. Consider the following data table::

|      |          |          |         |
|------|----------|----------|---------|
| x    | 1.1      | 1.2      | 1.3     |
| f(x) | -0.57941 | -0.90730 | -1.2807 |

- Using the above data, compute  $f'(1.2)$  using the central difference method.
- For the interval  $[1.1, 1.3]$ , compute the error bound (truncation error) if the above data is generated by the function,  $f(x) = x \cos(x) - x^2 \sin(x)$ .
- Also compute the actual error.